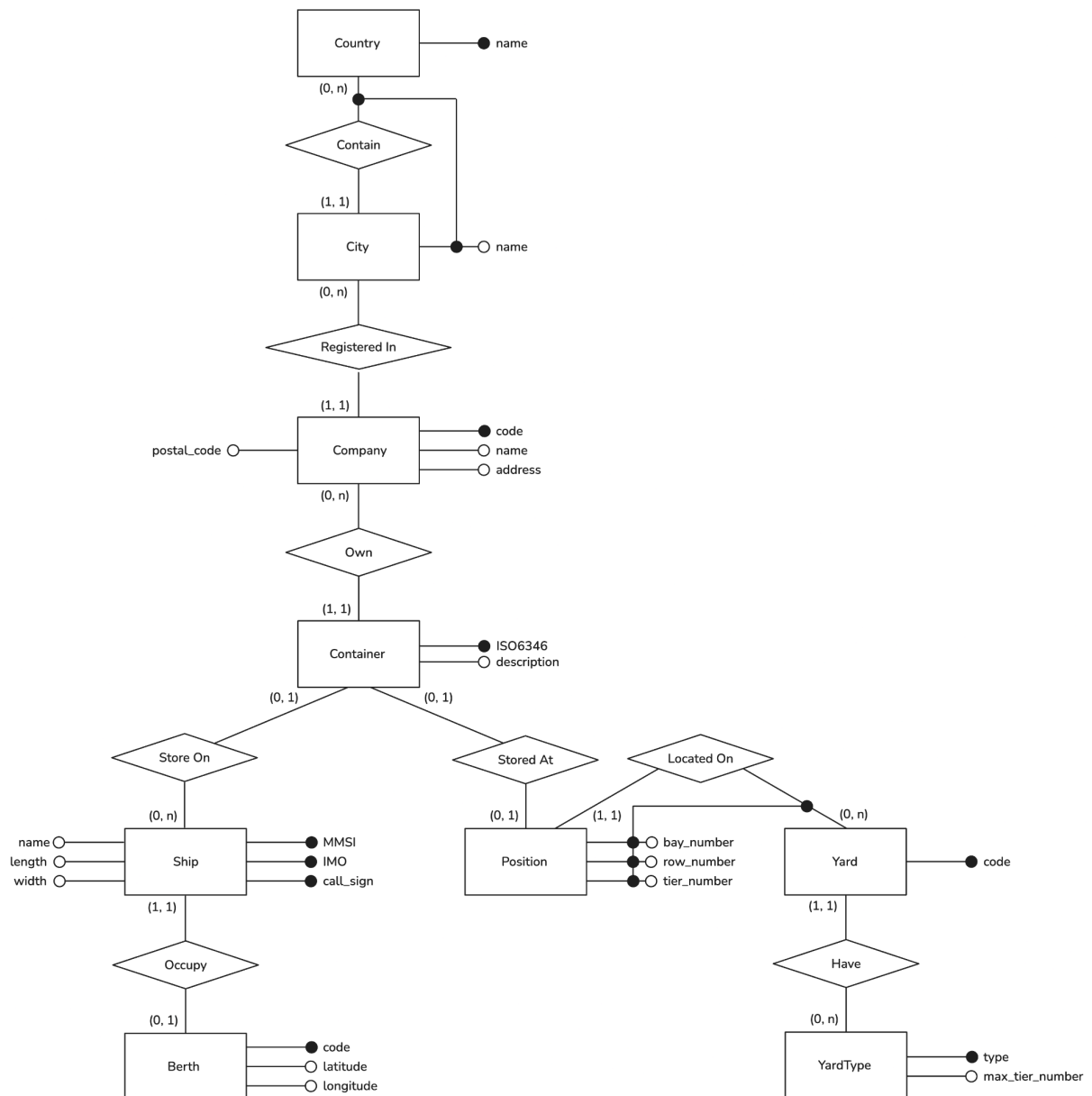


CS2102 Project 2: Entity-Relationship Diagram

Group Number: 77

Entity-Relationship Diagram



Design Decisions - Entities & Relationships

- **Country (0, n) - Contain - (1, 1) City:**
A country may have many cities or none, and a city belongs to exactly one country. This allows countries with no recorded cities.
- **Company (1, 1) - Registered In - (0, n) City:**
A city may have any number of companies or none, and a company is registered in exactly one city. A company's country is reached through its city, so it is not stored again.
- **Company (0, n) - Own - (1, 1) Container:**
A company may own no containers, and every container has exactly one owner.
- **Ship (1, 1) - Occupy - (0, 1) Berth:**
Every recorded ship occupies exactly one berth, and a berth holds at most one ship. A berth may be empty but must still be recorded.
- **Container (0, 1) - Store On - (0, n) Ship:**
A container is on at most one ship, and a ship may carry any number of containers.
- **Container (0, 1) - Stored At - (0, 1) Position:**
A container occupies at most one position, and a position holds at most one container. A position may be empty, which supports finding unused positions.
- **Position (1, 1) - Located On - (0, n) Yard:**
Every position belongs to exactly one yard. A yard may have no positions yet, and so no containers.
- **Yard (1, 1) - Have - (0, n) YardType:**
Every yard has exactly one type, and a type may have no yards, such as an AI-enabled type not yet in use.

Design Decisions - Entity Keys and Modelling Choices

- **Company** is identified by **code**, which is the three-letter prefix of its containers' ISO 6346 codes. **name** is not an identifier because company names are not guaranteed to be unique. **address** and **postal_code** are ordinary attributes.
- **City** is a weak entity identified by (**Country.name**, **City.name**) through the identifying relationship **Contain**, because cities in different countries can share a name (the 52 San Antonios).
- **Ship** has three candidate keys, **MMSI**, **IMO** and **call_sign**, because a ship can be identified by any of the three.

- **Berth** is identified by **code**, with **latitude** and **longitude** as attributes.
- **Position** is a separate weak entity identified by (**Yard.code**, **bay_number**, **row_number**, **tier_number**). A yard can exist with no positions, and a position exists whether or not it holds a container.
- **Yard** is identified by **code**. Its type and its positions are captured by the **Have** and **Located On** relationships, so it has no other attributes.
- **YardType** is a separate entity identified by **type**, with **max_tier_number** as an attribute. The limit belongs to the type, can change when the port is upgraded, and a type may have no yards.
- **Container** is identified by **ISO6346**, with **description** as an attribute.

Constraints Not Captured by the ER Diagram

- It is impossible to represent a **XOR relationship** (relate to one and only one of two optional parent entities, never both or neither) with an ERD diagram: it is possible to have a container stored in both a **Position** in a yard and on a **Ship**, or in neither of them.
- The diagram does not enforce that **tier_number** does not exceed the **max_tier_number** allowed by the **YardType** of the **Yard** that the **Position** is on.
- The diagram does not enforce valid stacking. A container at a **tier_number** higher than 1 should not exist without a container at the tier directly beneath it in the same yard, bay, and row.
- The diagram does not enforce natural-number restrictions for berth codes, bay numbers, row numbers, and tier numbers.
- The diagram does not enforce valid ISO 6346, MMSI, IMO, or call-sign formats.
- The diagram does not enforce that the first three letters of a container's **ISO6346** identifier match the **code** of its owning **Company**.
- The diagram does not enforce valid ship dimensions (positive **length** and **width**), postal-code formats, or valid ranges and precision for berth **latitude** and **longitude**.
- The diagram assumes that city names are unique within one country, since **City** is identified by (**Country.name**, **City.name**). Two different cities with the same name in the same country cannot be recorded.